COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 / Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	7/13/26	**DATE SUBMITTED:**	7/13/26
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ACTIVITY NO. 1.1 Basic C++ Programming

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A: In here, the program uses function prototypes and calls each function in main(). Each function performs a specific task like adding two numbers, comparing two values, and displaying logical operations

```cpp
#include <iostream>

void displaySum(int a, int b);
bool isGreater(int A, int B);
bool logicalOperations(bool x, bool y);

int main()
{
    displaySum(10, 15);

    if (isGreater(20, 10))
        std::cout << "A is greater than B.\n";
    else
        std::cout << "A is NOT greater than B.\n";

    if (logicalOperations(true, false))
        std::cout << "Logical operations executed successfully.\n";

    return 0;
}

void displaySum(int a, int b)
{
    std::cout << "Sum = " << a + b << std::endl;
}

bool isGreater(int A, int B)
{
    return (A > B);
}

bool logicalOperations(bool x, bool y)
{
    std::cout << "AND: " << (x && y) << std::endl;
    std::cout << "OR : " << (x || y) << std::endl;
    std::cout << "NOT x: " << (!x) << std::endl;
    std::cout << "NOT y: " << (!y) << std::endl;

    return true;
}
```

Output:

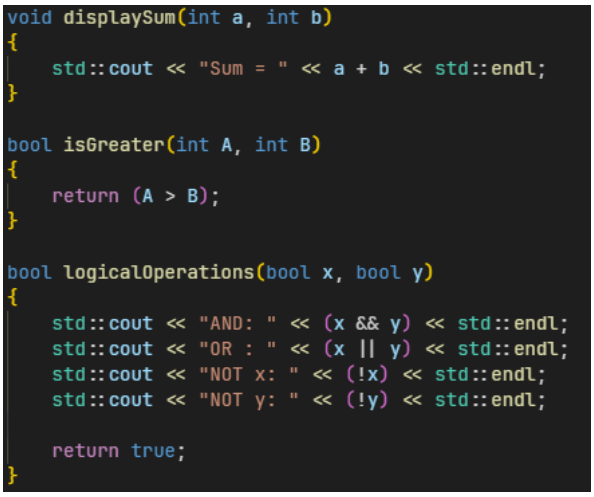
```
Sum = 25
A is greater than B.
AND: 0
OR : 1
NOT x: 0
NOT y: 1
Logical operations executed successfully.
```

Section	Answer
Header File Declaration Section	`#include <iostream>`

Global Declaration Section	`void displaySum(int a, int b);`
	`bool isGreater(int A, int B);`
	`bool logicalOperations(bool x, bool y);`

Class Declaration and Method Definition Section	

Main Function	`int main()`
	`{`
	`  displaySum(10, 15);`
	`  if (isGreater(20, 10))`
	`  std::cout << "A is greater than B.\n";`
	`  else`
	`  std::cout << "A is NOT greater than B.\n";`
	`  if (logicalOperations(true, false))`
	`  std::cout << "Logical operations executed successfully.\n";`
	`  return 0;`
	`}`

Method Definition	

```
void displaySum(int a, int b)
{
    std::cout << "Sum = " << a + b << std::endl;
}

bool isGreater(int A, int B)
{
    return (A > B);
}

bool logicalOperations(bool x, bool y)
{
    std::cout << "AND: " << (x && y) << std::endl;
    std::cout << "OR : " << (x || y) << std::endl;
    std::cout << "NOT x: " << (!x) << std::endl;
    std::cout << "NOT y: " << (!y) << std::endl;

    return true;
}
```

ILO B: In here it creates a Triangle object using a constructor that stores the three angles. The validateTriangle() function checks if their sum is 180°. Since the angles total 180°, the program displays that the shape is a valid triangle.

```cpp
#include <iostream>

class Triangle
{
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    bool validateTriangle();
};

int main()
{
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle())
        std::cout << "The shape is a valid triangle.\n";
    else
        std::cout << "The shape is NOT a valid triangle.\n";

    return 0;
}

Triangle::Triangle(double A, double B, double C)
{
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

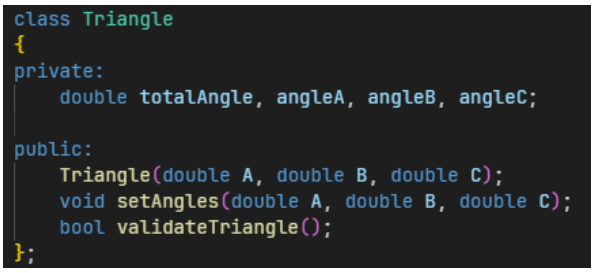
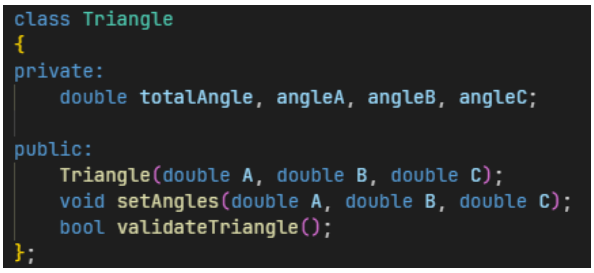
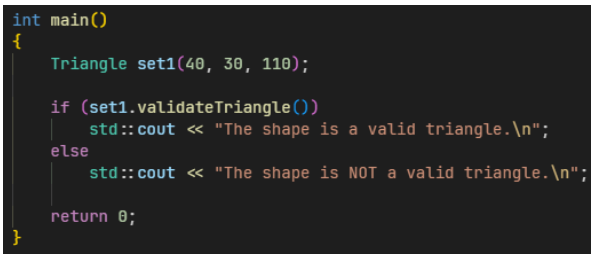
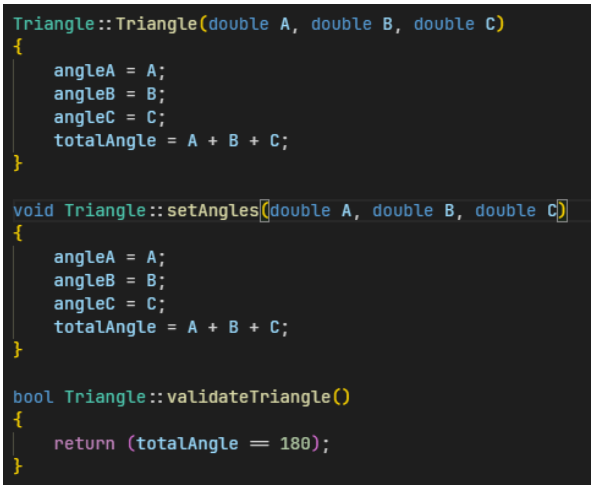
void Triangle::setAngles(double A, double B, double C)
{
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

bool Triangle::validateTriangle()
{
    return (totalAngle == 180);
}
```

Output:

```
The shape is a valid triangle.
PS D:\Space\Files\Jepi\ILO B>
```

Section	Answer
Header File Declaration Section	`#include <iostream>`

Global Declaration Section	
Class Declaration and Method Definition Section	
Main Function	
Method Definition	

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C:

1. It just accepts two numbers from the user and swaps their values using a temporary variable. It then displays the swapped values, demonstrating the use of variables, assignments, and input/output operations.

```cpp
#include <iostream>

int main()
{
    int a, b, temp;

    std::cout << "Enter first number: ";
    std::cin >> a;

    std::cout << "Enter second number: ";
    std::cin >> b;

    temp = a;
    a = b;
    b = temp;

    std::cout << "\nAfter swapping:\n";
    std::cout << "First number: " << a << std::endl;
    std::cout << "Second number: " << b << std::endl;

    return 0;
}
```

Output:

```
Enter first number: 2
Enter second number: 1

After swapping:
First number: 1
Second number: 2
```

2. It uses a function to convert a temperature from Kelvin to Fahrenheit. The user enters a Kelvin value, the function performs the conversion, and the result is displayed, showing how functions can be used to perform calculations.

```cpp
#include <iostream>

double kelvinToFahrenheit(double kelvin);

int main()
{
    double kelvin;

    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> kelvin;

    std::cout << "Temperature in Fahrenheit: "
              << kelvinToFahrenheit(kelvin) << std::endl;

    return 0;
}

double kelvinToFahrenheit(double kelvin)
{
    return (kelvin - 273.15) * 9 / 5 + 32;
}
```

Output:

```
Enter temperature in Kelvin: 20
Temperature in Fahrenheit: -423.67
```

3. It accepts the coordinates of two points and uses a function to calculate the distance between them using the distance formula. This demonstrates the use of mathematical functions and passing multiple parameters to a function.

```cpp
#include <iostream>
#include <cmath>

double distance(double x1, double y1, double x2, double y2);

int main()
{
    double x1, y1, x2, y2;

    std::cout << "Enter x1 y1: ";
    std::cin >> x1 >> y1;

    std::cout << "Enter x2 y2: ";
    std::cin >> x2 >> y2;

    std::cout << "Distance = "
              << distance(x1, y1, x2, y2)
              << std::endl;

    return 0;
}

double distance(double x1, double y1, double x2, double y2)
{
    return std::sqrt((x2 - x1) * (x2 - x1) +
                     (y2 - y1) * (y2 - y1));
}
```

Output:

```
Enter x1 y1: 2 3
Enter x2 y2: 6 6
Distance = 5
```

4. This modified the Triangle class that computes the triangle's perimeter and area and determines whether it is acute, obtuse, or right-angled. It reinforces the use of classes, objects, constructors, and member functions to solve a more complex problem.

```cpp
#include <iostream>
#include <cmath>

class Triangle
{
private:
    double sideA, sideB, sideC;

public:
    Triangle(double A, double B, double C);

    void setSides(double A, double B, double C);

    double getPerimeter();

    double getArea();

    void triangleType();
};

Triangle::Triangle(double A, double B, double C)
{
    sideA = A;
    sideB = B;
    sideC = C;
}

void Triangle::setSides(double A, double B, double C)
{
    sideA = A;
    sideB = B;
    sideC = C;
}

double Triangle::getPerimeter()
{
    return sideA + sideB + sideC;
}

double Triangle::getArea()
{
    double s = getPerimeter() / 2;

    return std::sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}
```

```cpp
void Triangle::triangleType()
{
    double a2 = sideA * sideA;
    double b2 = sideB * sideB;
    double c2 = sideC * sideC;

    if (a2 > b2)
    {
        double temp = a2;
        a2 = b2;
        b2 = temp;
    }

    if (b2 > c2)
    {
        double temp = b2;
        b2 = c2;
        c2 = temp;
    }

    if (a2 > b2)
    {
        double temp = a2;
        a2 = b2;
        b2 = temp;
    }

    if (a2 + b2 > c2)
        std::cout << "Triangle Type: Acute Triangle\n";
    else if (a2 + b2 < c2)
        std::cout << "Triangle Type: Obtuse Triangle\n";
    else
        std::cout << "Triangle Type: Others (Right Triangle)\n";
}

int main()
{
    Triangle t(3, 4, 5);

    std::cout << "Perimeter: " << t.getPerimeter() << std::endl;
    std::cout << "Area: " << t.getArea() << std::endl;

    t.triangleType();

    return 0;
}
```

Output:

```
Perimeter: 12
Area: 6
Triangle Type: Others (Right Triangle)
```

Summary of Lessons Learned:

This activity helped me understand the basic structure of a C++ program, including header files, class declarations, method definitions, and the main function. I also learned how to create classes, constructors, and member functions using object oriented programming principles but not that good.

Concluding Statement / Feedback:

I think I did not do well in this activity but I was able to complete the required programs and understand the basic concepts. The activities helped me improve my skills in using functions, classes, objects, and problem-solving techniques. My areas for improvement are writing more efficient and organized code, strengthening my debugging skills, and practicing more complex C++ programs to gain confidence and proficiency.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Quillbot, ChatGpt**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction, Code assistance, and Explanation of the concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **It just fix my grammar and make it better but it did not replace the original thinking It's my own work and ChatGpt for like how to use the function properly in the concept of cmath library**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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